

Stat 150– Lecture 01

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1 Markov Chain

Markov Chain: Current state is enough information

$$\mathbb{P}(X_{t+1} = a_{t+1} | X_t = a_t, X_{t-1} = a_{t-1}, \dots, X_0 = a_0) = P(X_{t+1} = a_{t+1} | X_t = a_t)$$

Random walk with absorbing states is not too hard, just solve lots of equations

1.1 Reflecting Random walk on a line graph

Suppose we have a random walk X_t takes values on $S = 0, 1, 2, \dots, N$. such that

$$P(i, i+1) = p, P(i, i-1) = 1-p, P(N, N-1) = P(0, 1) = 1 \text{ and } 0 < p < 1$$

In this case, at the previously absorbing states 0 and 1, they "reflect" so we don't get stuck and with probability 1 we always return to any state

1.2 Infinite Line Graph

Consider the previous example except $S = \mathbb{Z}$.

Let X_t describe the position at time t of the random walk.

In this random walk, do we visit any state infinitely many times?

Answer: Yes, if and only if $p = 1 - p = \frac{1}{2}$ and the walk is symmetric.

Extension: How does this work in $\mathbb{Z}^2$ or $\mathbb{Z}^3$??

Answer to Extension: $\mathbb{Z}^2$ if symmetric, $\mathbb{Z}^3$ hell no! (transient random walk)

1.3 Random walk over a circle

The transition matrix is

$$P = \begin{pmatrix} 0 & p & 0 & \cdots & 0 & q \\ q & 0 & p & \ddots & \vdots & 0 \\ 0 & q & 0 & \ddots & 0 & \vdots \\ \vdots & \ddots & \ddots & \ddots & p & 0 \\ 0 & \cdots & 0 & q & 0 & p \\ p & 0 & \cdots & 0 & q & 0 \end{pmatrix}_{(N+1) \times (N+1)}, \quad p + q = 1.$$

2 definitions and notation

Definition 2.1 (First definition). Write the definition and one sentence explaining why it matters.

3 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

4 Examples and calculations

Example 4.1 (Lecture example). Write the setup, the calculation, and the conclusion.

5 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: